

Laboratory assignment

Component (2)

Authors: Purdea Bogdan-Florin, Todoran Ciprian Bogdan

Group: ICA2

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1 Problem Definition

The main objective of this project is to simulate and analyze the outcomes of the **Prisoner's Dilemma** game. This game originates from the field of **Game Theory** and models the interaction between two rational agents, each having the option to either cooperate (*Cooperate*) or betray the other (*Defect*). The Prisoner's Dilemma represents a fundamental paradox in which two agents, acting in their own self-interest, may fail to achieve the optimal collective outcome.

Consider two players, denoted as *Red* and *Blue*. Each player independently chooses one of the two possible actions: *Cooperate* or *Defect*. The resulting payoffs are defined as follows:

- If both players cooperate, each receives the reward payoff R .
- If both players defect, each receives the punishment payoff P .
- If Blue defects while Red cooperates, Blue receives the temptation payoff T , while Red receives the sucker's payoff S .
- If Blue cooperates while Red defects, Blue receives the sucker's payoff S , while Red receives the temptation payoff T .

The objective of the project is to simulate a tournament in which multiple agents, each implementing a distinct strategy, interact repeatedly in pairwise matches. The purpose is to evaluate and identify the most effective strategy in terms of accumulated payoff and behavioral performance.

The payoff structure must satisfy the condition:

$$T > R > P > S$$

The relation $R > P$ indicates that mutual cooperation yields a better outcome than mutual defection. However, the inequalities $T > R$ and $P > S$ imply that defection is the dominant strategy for both agents, leading to the classical dilemma.

	Blue: Cooperate	Blue: Defect
Red: Cooperate	(5, 5)	(0, 10)
Red: Defect	(10, 0)	(1, 1)

In this matrix, each ordered pair represents the payoff (Red, Blue). The values correspond to the following:

- $T = 10$ (Temptation to defect);
- $R = 5$ (Reward for mutual cooperation);
- $P = 1$ (Punishment for mutual defection);
- $S = 0$ (Sucker's payoff).

2 Multi-Agent System Specification

To address the problem, a Multi-Agent System (MAS) is designed and implemented using the **SPADE** framework. The system models an iterated version of the Prisoner's Dilemma, where autonomous agents interact repeatedly over a fixed number of rounds.

2.0.1 Inputs and Outputs of the MAS

Inputs:

- Number of rounds per match (e.g., 1000 iterations);
- Payoff matrix parameters: T , R , P , S , satisfying $T > R > P > S$;
- Set of participating player agents;
- Strategy assigned to each player agent.

Outputs:

- Round-level interaction data (selected actions and resulting payoffs);
- Final cumulative scores for each agent;
- Aggregated performance metrics (e.g., cooperation rate, total payoff);
- Persistent storage of results in *.CSV* format for further analysis.

2.0.2 Types of Agents

The system consists of two categories of agents:

- **Manager Agent:** responsible for coordination, control and data management;
- **Player Agents:** autonomous entities implementing decision-making strategies.

2.0.3 Agent Specification

Manager Agent

Inputs:

- Actions received from player agents;
- Simulation configuration parameters.

Outputs:

- Action requests sent to player agents;
- Round results, including opponent actions and computed payoffs;
- Final match summaries.

Tasks:

- Initialize and manage matches between player agents;
- Enforce synchronization across simulation rounds;
- Collect player actions and compute corresponding payoffs;
- Maintain the global state of the simulation;
- Persist results to *.CSV* storage.

Player Agents

Inputs:

- Action requests from the Manager Agent;
- Information regarding previous interactions (e.g., opponent's last action).

Outputs:

- Selected action: *Cooperate* or *Defect*.

Tasks:

- Execute a predefined decision-making strategy;
- Maintain an internal history of interactions;
- Determine actions based on strategy and historical data.

2.0.4 Communication Between Agents

Communication between agents is implemented using the XMPP-based messaging infrastructure provided by the SPADE framework. The system follows an asynchronous message-passing model.

The interaction protocol proceeds as follows:

1. The Manager Agent sends an action request to each Player Agent at the beginning of a round;

2. Each Player Agent responds with its selected action (*Cooperate* or *Defect*);
3. The Manager Agent computes the resulting payoffs using the predefined payoff matrix;
4. The Manager Agent communicates the round outcome to both Player Agents.

This sequence is repeated for a predefined number of rounds, ensuring proper synchronization and consistency of the simulation.

2.0.5 Agents' Roles within the MAS

- **Manager Agent:**

- Acts as the central coordinator of the system;
- Ensures correct execution order and synchronization;
- Maintains global knowledge of the simulation;
- Handles data persistence and result aggregation.

- **Player Agents:**

- Represent autonomous rational decision-makers;
- Implement distinct behavioral strategies;
- Operate based on local knowledge and interaction history;
- Adapt decisions according to observed outcomes;